

UNIVERSITY OF PERADENIYA

**POWER GENERATION WITH FOOTSTEPS OF
STAIRS AND USES OF FOOTSTEPS FOR
ADVANCE SECURITY PURPOSES**

By

MEEGAHAKOTUWA T. B. (E/19/239)

RAJAPAKSHE R. H. T. M. (E/19/308)

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Supervised by:
Dr. Janaka Rajaguru

Department of Mechanical Engineering
Faculty of Engineering

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Lastly, we aim to improve sustainable energy production and incorporate advanced security measures for the benefit of society.

Abstract

Power generation through the actuation of a staircase involves a kinetic-to-electrical energy transduction system, utilizing a piston-cylinder mechanism coupled with a spring-mass system beneath each stair tread. When incorporating renewable energy generation technologies, cost-effectiveness relative to existing methods remains a primary concern. Energy harvesting from human motion, while a developing field, presents significant untapped potential. The scope of this project is to harvest ambient kinetic energy, typically dissipated as waste during routine pedestrian activities such as walking. This proposed system incorporates two primary features: first, the aforementioned localized energy generation; and second, biometric human identification utilizing footstep pressure and acceleration patterns. As pedestrians ascend or descend the staircase, integrated vibration and acceleration sensors capture dynamic motion characteristics. This sensor data is subsequently analyzed to detect unique gait and walking patterns for each individual. The data collection and preprocessing pipeline feeds into an Artificial Neural Network (ANN) designed to accurately classify and identify individuals based on their biometric footstep signatures.

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Chapter 1

Introduction

1.1 Background Introduction

Conventional power producing techniques are under tremendous strain due to the world's energy demand's explosive development. In addition to contributing significantly to environmental pollution, including greenhouse gas emissions and air contamination, conventional systems mostly rely on fossil fuels like coal, oil, and natural gas, which deplete finite natural resources. As crucial elements of the global effort to fight climate change and protect ecosystems, renewable energy sources and ecologically friendly electricity generation techniques are now receiving more attention as a result of these problems.

Even while massive renewable energy projects like wind and solar farms have become more well-known, localized and small-scale electricity production has also become a practical way to meet certain energy demands. These systems are especially useful in environments with low electricity demand and abundant natural resources or unexplored potential. The kinetic energy produced by human movement during routine activities like walking, dancing, running, or even working out is one such possibility. Despite being plentiful, this type of energy is frequently lost and releases into the environment without being used.

Recent technological developments have produced techniques and apparatuses that can capture and transform human kinetic energy into electrical power that can be used. Through the use of piezoelectric materials, electromagnetic induction, and other mechanisms, these techniques harvest energy from daily activities. Even though these systems have shown promise in producing sustainable energy, there is still much room for innovation in terms of maximizing their effectiveness, usability, and flexibility.

By suggesting the creation of a novel power generation system that combines human kinetic energy harvesting with a person identification method, this initiative advances the field. In order to power small devices or systems, the idea seeks to harvest and transform the kinetic energy generated during human activities into electrical energy. The system can also provide customized functionalities, enhance security, and provide individualized energy usage tracking by integrating a person identification component. A cleaner and more energy-efficient future is made possible by this combination of energy harvesting and cutting-edge identification technology, which is in line with current trends in smart systems, sustainable design, and renewable energy use.

1.2 Problem Statement

For safety and functionality, staircase lighting is crucial, especially in public areas, business buildings, and residential buildings. The most energy-efficient lighting choice is LEDs, which can use as low as 5W to 10W while still producing illumination that is on par with or better. The total energy consumption for a stairway with ten lights, each using 10W, and running for ten hours per day is about 1 kWh per day or 30 kWh per month. Systems using renewable energy drastically save electricity costs for lighting stairwells.

1.3 Future scope of power generation by human walking

Walking energy can be used to power or enhance devices like smartwatches, fitness trackers, and medical monitoring systems. Its battery life can be improved by lessening reliance on external power sources. Electricity can be produced from foot movement via kinetic energy-harvesting systems installed in public areas' pavements, walkways, or flooring. Streetlights in the area might be powered by harvested energy, improving sustainability and lowering dependency on the grid.

Chapter 2

Literature Review

As a sustainable energy option, power generation through footsteps using spring-based systems has drawn a lot of attention. This method transforms the kinetic energy produced by human steps into electrical energy by utilizing mechanical parts including plates, springs, rods, and alternators. Motion is transferred to a linked rod when a person steps on the plate because the downward force compresses the spring. An alternator is then powered by this rod to generate electricity. This style is well-known for being straightforward, effective, and useful, which makes it appropriate for a range of uses.

The spring mechanism's capacity to increase the energy produced by each step is one of its main advantages. A more effective transformation of mechanical energy into electrical energy is ensured by the spring's compression and subsequent release, which increases the force sent to the alternator. Furthermore, springs and rods' dependability and durability make these systems perfect for prolonged use in high-traffic areas.

Such systems have a wide range of possible uses, from integration into public infrastructure to small-scale power generation for gadgets like LED lights and sensors. Because of their heavy foot traffic, urban places like train stations, airports, and sidewalks are especially well-suited for this technology, which maximizes energy output. Energy-harvesting tiles and panels in public areas, such as Boulder Plaza in Las Vegas, are notable examples of comparable concepts in action. The energy produced there powers the lamps in the area. These uses highlight how feasible it is to produce energy locally by using human kinetic energy.

Research has also looked into ways to improve these systems' efficacy and efficiency. For example, energy conversion rates have increased as a result of improving the materials used in alternators and springs. According to research, even though each step uses a small amount of energy, the combined impact of several users in a crowded setting can greatly lessen need on conventional energy sources.

Despite its benefits, there remain obstacles to go beyond. Due to the intermittent nature of footstep energy, efficient energy storage is still a major challenge. To store and use the power effectively, reliable technologies like batteries or capacitors are needed. Furthermore, careful design and material optimization are required for expanding the technology for larger systems.

However, new developments in the integration of these systems with IoT devices and smart technologies offer promising prospects for further growth. To sum up, power production using spring-based footstep mechanisms is a promising area that blends useful applications with mechanical efficiency. These devices support sustainability

objectives and renewable energy projects by harnessing the kinetic energy of human movement, especially in metropolitan environments. The scalability and incorporation of such systems into regular infrastructure will probably increase their usefulness and influence as technology develops.

Chapter 3

Methodology

3.1 Introduction

Understanding the fundamentals of kinetic energy-based power generation is a necessary first step. A hydraulic piston may transform the kinetic energy of a footstep into pressure, and the fluid flow can transform the pressure exerted on it into velocity and pressure. Rotational velocity is the result of that energy conversion. Using a DC generator, mechanical energy can be transformed into electrical power. When needed, the produced energy is stored in battery packs. Flow diagram for setup is shown below:

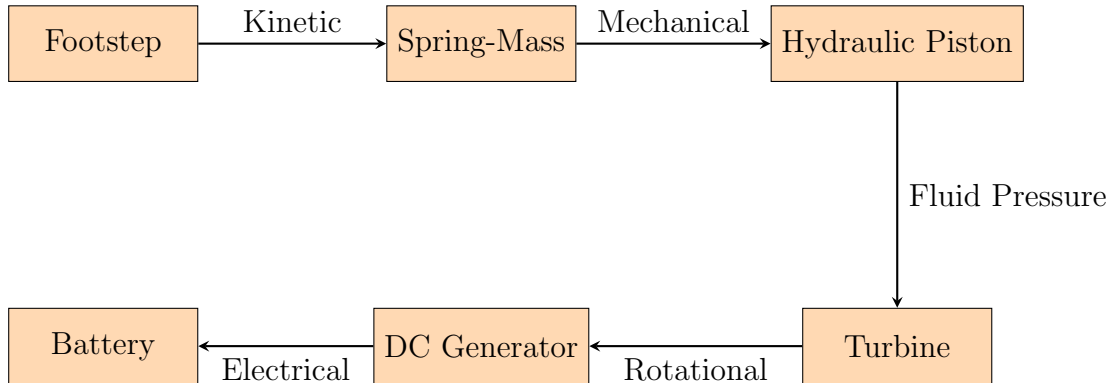


Figure 3.1: Energy Conversion Flow Diagram

3.2 Calculations

Height of stair taken as 25cm and maximum displacement induced by 100kg weighted person taken as 5cm.

Spring size calculations:

1. Force-Deflection Relationship (Hooke's Law)

The force exerted by or on a spring is proportional to its deflection:

$$F = k \cdot x \quad (3.1)$$

$$100kg \cdot 9.81 m/s^2 = k \cdot 0.05m \quad (3.2)$$

$$k = 19620 N/m \quad (3.3)$$

Where:

- F : Force applied to the spring (N)
- k : Spring constant or stiffness (N/m)
- x : Displacement or compression/stretch of the spring (m)

2. Spring Constant (Stiffness)

For a helical spring, the spring constant k is calculated using:

$$k = \frac{Gd^4}{8D^3n} \quad (3.4)$$

Where:

- G : Shear modulus of the spring material (N/m^2) = 80 GN/m^2 . Material for spring is structural steel.
- d : Diameter of the spring wire (m) = 0.004 m
- n : Number of active coils (turns)
- D : Mean diameter of the spring (m) = 0.04 m

By substituting the above values into the equation:

$$n = 20 \quad (3.5)$$

3. Spring Size Parameters

To determine the size of a spring, consider the following:

1. **Wire Diameter (d):** The thickness of the wire affects the spring's stiffness and maximum load capacity.
2. **Mean Diameter (D):** This is the average of the inner and outer diameters of the spring:
$$D = \frac{D_{outer} + D_{inner}}{2} \quad (3.6)$$
3. **Number of Coils (n):** The number of active coils influences the spring's flexibility and maximum extension/compression.
4. **Free Length (L):** The length of the spring when no force is applied.
5. **Maximum Deflection (ΔL_{max}):** The maximum distance the spring can compress or extend safely.

3.2.1 Hydraulic Power Transfer

The mechanical energy stored in the spring and exerted by the footstep is transferred to a hydraulic fluid via a double-acting piston-cylinder arrangement. The mechanical work (W_{mech}) done by the piston is given by:

$$W_{mech} = \int F dx = F \cdot \Delta x \quad (3.7)$$

Where F is the average force applied and Δx is the piston stroke.

This work creates fluid pressure. The hydraulic power (P_{fluid}) transferred to the turbine can be expressed as:

$$P_{fluid} = \Delta P \cdot Q \quad (3.8)$$

Where:

- ΔP : Pressure difference across the turbine (Pa)
- Q : Volumetric flow rate of the fluid (m^3/s)

3.2.2 Electrical Power Generation

The high-pressure fluid drives a micro-turbine coupled with a DC Generator. The mechanical power of the turbine (P_{turb}) is converted to electrical power (P_{elec}). The electrical power generated is given by:

$$P_{elec} = V \cdot I = P_{turb} \cdot \eta_{gen} = \tau \cdot \omega \cdot \eta_{gen} \quad (3.9)$$

Where:

- V : Generated voltage (V)
- I : Current generated (A)
- τ : Torque applied to the generator shaft (Nm)
- ω : Angular velocity of the generator shaft (rad/s)
- η_{gen} : Efficiency of the generator system

3.3 Computer Aided Designs

This chapter focus about Computer Aided Design of proposed model using SolidWorks software package and stress strain analysis with ANSYS software packages. Designed mechanism for the power generation setup is shown in the figures below:

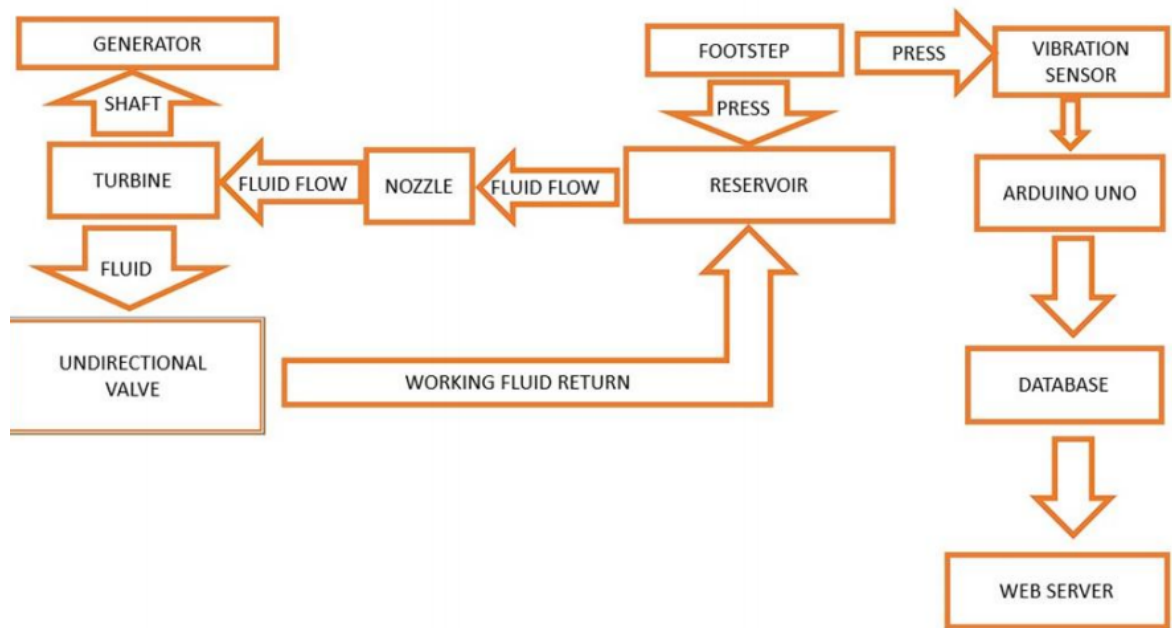


Figure 3.2: Flow Diagram

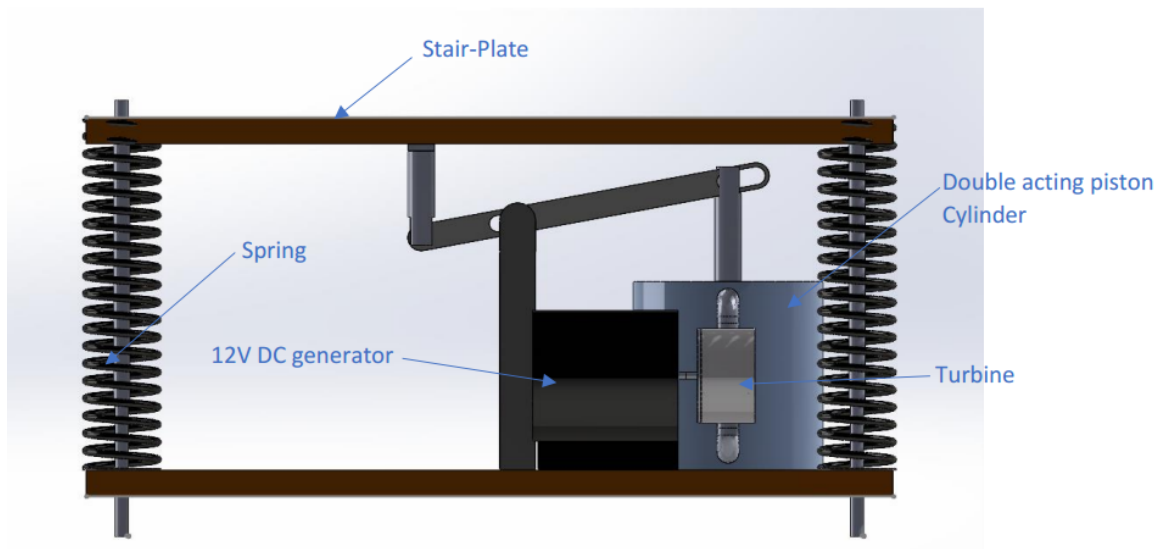


Figure 3.3: 3D model



Figure 3.4: Turbine

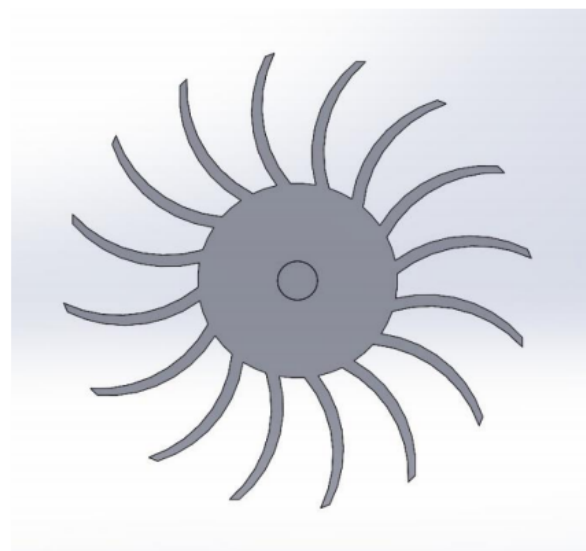


Figure 3.5: Sectional View of Turbine



Figure 3.6: Double acting Piston Cylinder Arrangement

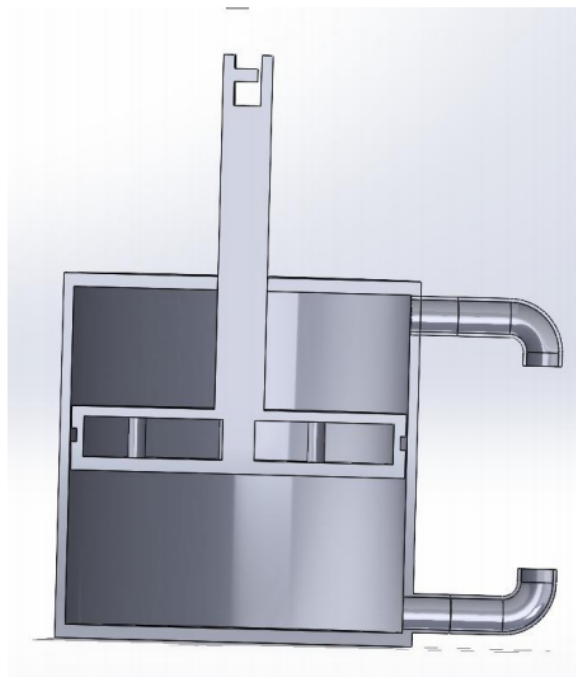


Figure 3.7: Sectional View of Double Acting Piston Cylinder



Figure 3.8: Lever Mechanism

Chapter 4

People Identification Using Walking Patterns

4.1 Introduction

Identifying people based on their unique walking patterns is an emerging field of biometric identification. Our project explores the use of vibration sensors to capture and analyze footsteps to distinguish people as they traverse staircases. Our system utilizes Firebase Cloud Storage for centralized data handling and a Flutter mobile application for visualization and predictions. This innovative approach demonstrates potential for walking patterns to serve as a reliable biometric identifier, with applications in security, smart homes, and health monitoring.

4.2 Objectives

The main goal of this research is to use cutting-edge technologies to analyze each person's distinct walking patterns in order to precisely identify them. The initial objective is to use vibration sensors to gather information about walking patterns. These sensors record important metrics that provide the basis for precise pattern recognition, including time spent per step and multidimensional x, y, and z accelerations.

Integrating the gathered data with Firebase Cloud Storage is our next goal. This guarantees that all data is safely kept and easily processed and retrieved. Firebase is a dependable platform for effectively managing massive amounts of sensor data because it also allows for real-time updates and centralized administration.

The creation of the mobile application with Flutter is another important goal. This application offers a user-friendly interface for displaying data on walking patterns and making identity predictions using the information gathered. This software gives users an intuitive experience by combining prediction and visualization features.

Ultimately, the application of a machine learning system results in the precise prediction of individual identities. The method highlights the potential of walking patterns as a biometric identifier by precisely identifying individuals through the analysis of walking behavior data.

4.3 Methodology

4.3.1 Data Collection

As people climbed and descended stairs, a vibration sensor was used to record specific footstep data. Numerous parameters necessary for examining walking patterns were recorded by this sensor. The duration of each step taken was indicated by the time spent each step, which was one of these factors. A depiction of the movement dynamics and differences in walking behavior was also obtained by collecting multidimensional acceleration data in the X, Y, and Z axes.

With each record identified to correspond to a certain user (a1, b1, etc.), the gathered data was prepared into a structured manner. By acting as the identifiers, these labeled records made it possible to accurately link the data to certain individuals. The data was saved in a CSV (Comma-Separated Values) format to facilitate simple access and additional processing. The project's predictive capabilities were built on this format, which made it easy to integrate with the machine learning model and other analytical tools.

4.3.2 Data Preprocessing

In order to prepare the raw sensor data for efficient machine learning and precise identification, the acquired data underwent preprocessing to guarantee its quality and usability for analysis and prediction. In order to preserve only significant walking patterns, the preprocessing included noise reduction, which required applying filters to remove background noise and unnecessary signals.

In order to reduce unpredictability brought on by negligible sensor fluctuations, feature standardization was also carried out by rounding numerical quantities, such as time spent each step and acceleration values (X, Y, Z), to three decimal places. Furthermore, every data record was tagged with distinct identities that corresponded to known persons. This was essential for supervised learning since it enabled the machine learning model to link particular users to walking patterns. This thorough pretreatment pipeline created a strong basis for precise prediction and user identification by guaranteeing the data's dependability and preparedness for ensuing machine learning applications.

4.3.3 Firebase Cloud Storage Integration

The footstep data was effectively stored, managed, and retrieved using Firebase Cloud Storage, which provides a stable and expandable framework for processing vibration sensor data. It operated as a central repository where all sensor data was uploaded, ensuring accessibility from a single, uniform place, thus simplifying management and retrieval processes.

In order to guarantee that the Flutter-based mobile application always displayed the most recent information, Firebase also made real-time synchronization possible. This allowed adjustments to the data to be instantaneously reflected in the application. Firebase offers safe storage solutions, combining built-in authentication and encryption algorithms to protect sensitive user data during storage and retrieval. This created a dependable backend for the mobile application by utilizing Firebase Cloud Storage, which guaranteed effective data processing, improved accessibility, and strong security.

4.3.4 Predictive Model

To identify individuals based on their complex, non-linear walking habits, the system employs an Artificial Neural Network (ANN), specifically a Multi-Layer Perceptron (MLP). The input features consist of the normalized time spent per step alongside the X, Y, and Z acceleration values. These features are fed into a sequential neural network architecture featuring multiple dense hidden layers utilizing Rectified Linear Unit (ReLU) activation functions to capture non-linear dynamic patterns in the gait.

The output layer employs a Softmax activation function to perform multi-class classification, outputting a probability distribution across all registered individuals in the dataset. The model is trained using categorical cross-entropy loss. Accuracy and an easy-to-use interface are guaranteed by the model's smooth integration via API to the Flutter-based mobile application, allowing real-time predictions of an individual's identity based on their latest footstep data.

4.3.5 Flutter Application

The main user interface is the mobile application, which offers data visualization and real-time forecasts. It gives customers a clear understanding of the data gathered by displaying the recorded parameters, such as time spent per step and X, Y, and Z acceleration values. The software has a prediction module that allows users to instantly identify people by either manually entering data or retrieving real-time data from sensors. The application's user-friendly interface, which prioritizes user experience, allows for easy navigation and smooth data interaction, guaranteeing efficiency and accessibility for a wide range of users.

4.4 Results

By successfully identifying people based on their distinct walking patterns, the system reached significant milestones. Multiple people's unique walking patterns were successfully gathered, creating a trustworthy dataset for study. The efficiency of the prediction model was validated when it showed a high degree of confidence in accurately identifying individual identities. Furthermore, the Flutter mobile application's real-time data retrieval and prediction features were smoothly incorporated to guarantee immediate and useful usability. Significant scalability and robustness in the model's performance demonstrated its promise for real-world uses in biometric identification and related domains.

4.5 Conclusion

This chapter offered a thorough method for identifying people based on their walking habits. The system provided a scalable and intuitive real-time prediction solution by combining vibration sensors, Firebase Cloud Storage, and a Flutter application. The research showed that employing walking patterns as a trustworthy biometric measure is feasible.

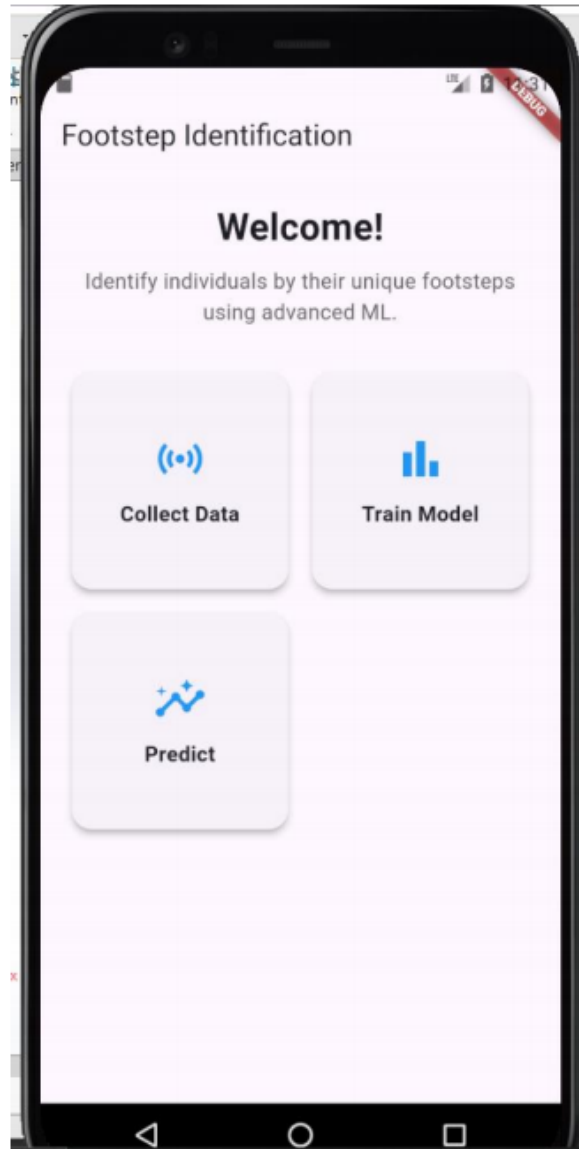


Figure 4.1: Main interface of API

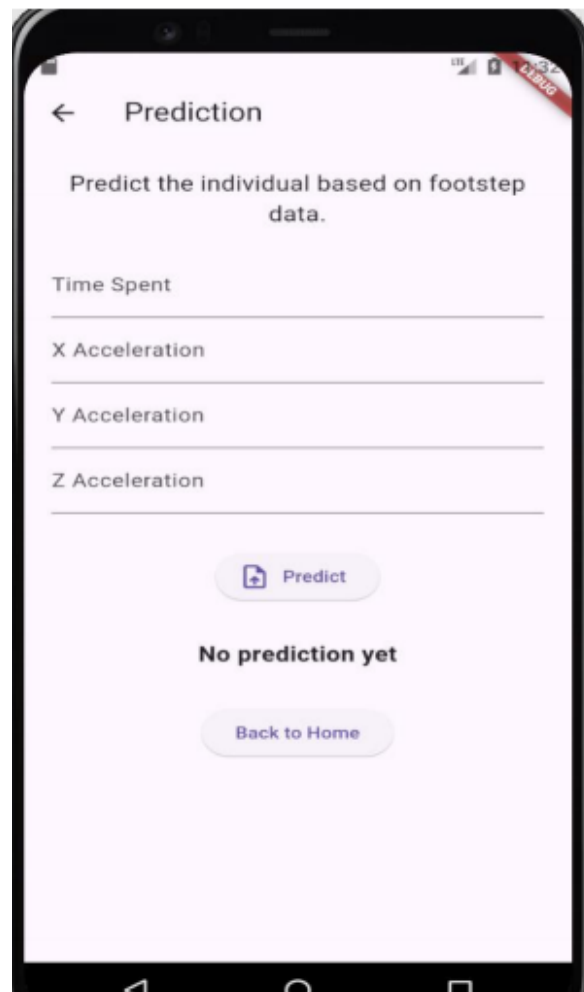


Figure 4.2: Prediction Page

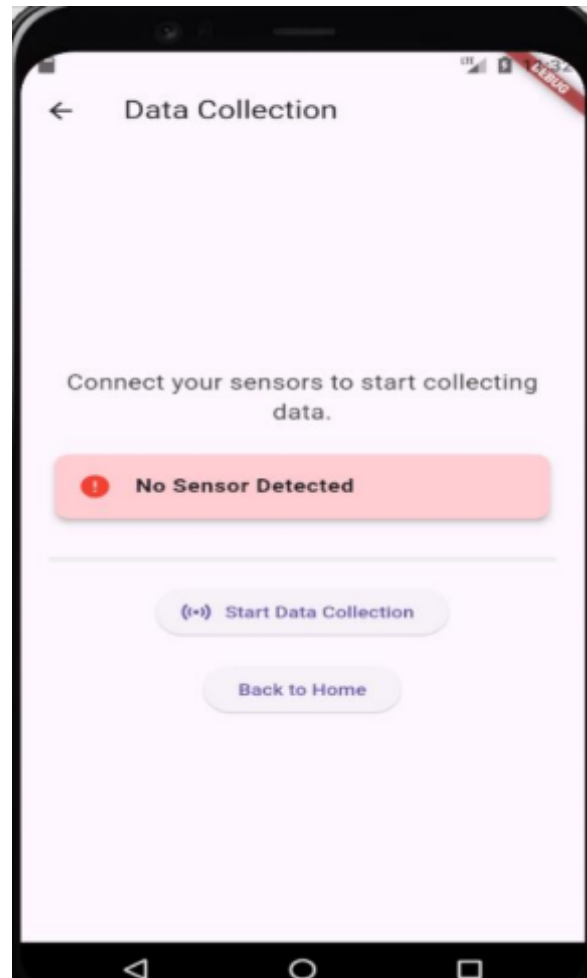


Figure 4.3: Data collection page

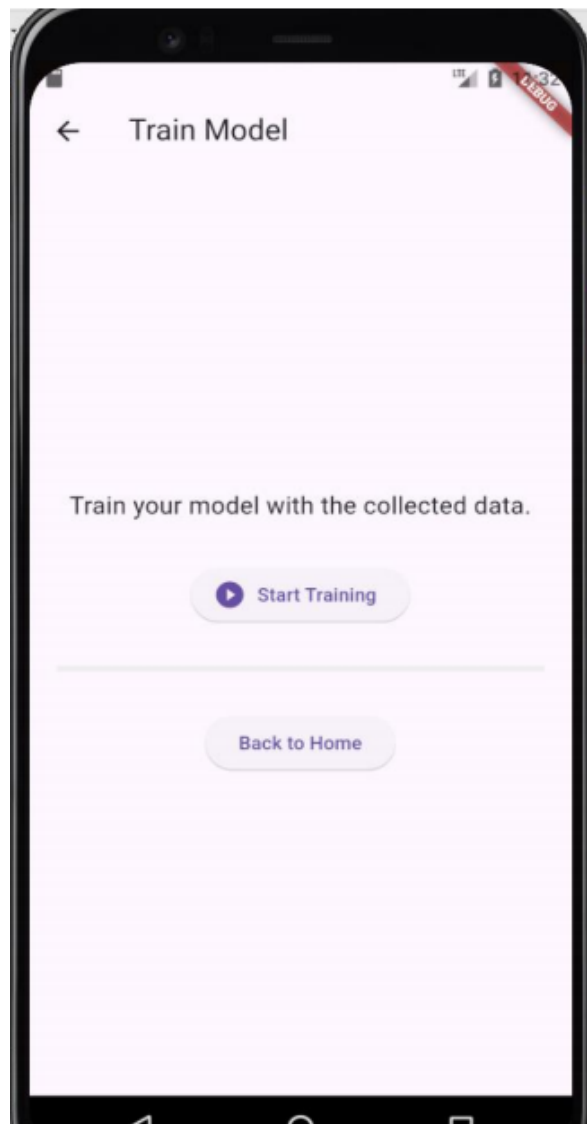


Figure 4.4: Data Training Page

4.6 Future Work

A number of enhancements are suggested to further improve the system. First, the system's capacity to manage intricate and subtle walking patterns may be improved by utilizing cutting-edge machine learning models, such as deep learning approaches, which would increase prediction accuracy. Second, by enabling on-device data analysis, edge processing would lessen need on cloud infrastructure and provide faster forecasts. Third, the system might be made more scalable by adding more sensors and processing bigger datasets, which would enable it to be used in a variety of settings and situations. Lastly, the system's capability would be expanded by merging it with wearable technology, which would allow portable sensors to record walking patterns in real-time and lead to more extensive applications in fields like security and health monitoring.

Appendix A

Python Code for ML Training

```
1 import numpy as np
2 import pandas as pd
3 from tensorflow import keras
4 from sklearn.model_selection import train_test_split
5 from sklearn.preprocessing import LabelEncoder, StandardScaler
6 import matplotlib.pyplot as plt
7
8 # Data preparation
9 data = pd.DataFrame({
10     'name': ['a1', 'a2', 'a3', 'a4', 'a5', 'a6', 'a7', 'a8', 'a9', 'a10', 'a11', 'a12', 'a13', 'a14', 'a15', 'a16', 'a17', 'a18', 'a19', 'a20', 'a21', 'a22', 'a23', 'a24', 'a25', 'a26', 'a27', 'a28', 'a29', 'a30', 'a31', 'a32', 'a33', 'a34', 'a35', 'a36', 'a37', 'a38', 'a39', 'a40', 'a41', 'a42', 'a43', 'a44', 'a45', 'a46', 'a47', 'a48', 'a49', 'a50'],
11     'x_acceleration': [0.69464, 0.14982, -0.29965, -1.6889, -2.1793, -2.3018, -1.4165, -0.27241, -0.61292, 1.3348, 3.4459, 1.4982, -1.7979, -2.3699, -1.4982, -0.34051, -0.23155, -0.53119, -1.5663, -1.5255, -0.88532, -1.3076, -0.65378, 0.23155, -0.42223, -0.57205, -0.65378, -0.34051, -0.34051, -0.84446, -1.0351, -1.1169, -1.2258, -1.0351, -0.8036, -0.72188, -1.185, -1.6072, -1.7298, -1.757, -1.757, -1.9205, -1.8387, -1.7298, -1.757, -1.757, -1.757, -1.9205, -1.757, -1.5663],
12     'y_acceleration': [3.1735, 3.4868, 1.9477, 1.4165, 0.95342, 0.23155, 1.185, 2.2201, 2.2201, 4.2495, 7.5865, 4.9033, 1.1169, 0.10896, 2.3427, 3.2144, 2.4925, 1.5391, 1.0351, 1.2258, 2.9148, 2.3699, 2.8739, 3.909, 2.6423, 2.4925, 3.4459, 3.5277, 3.1463, 3.2144, 3.1463, 2.8739, 3.5277, 3.5958, 3.7184, 3.8273, 3.4051, 3.4051, 3.4868, 3.6366, 3.9908, 4.1406, 4.0589, 3.7184, 3.8682, 3.7592, 3.6775, 3.8682, 4.0589, 4.1814],
13     'z_acceleration': [7.5048, 9.2755, 9.112, 10.12, 10.924, 10.651, 11.073, 11.986, 11.986, 11.414, 12.599, 10.692, 9.9156, 9.003, 9.2346, 8.2403, 8.4991, 9.1529, 10.801, 11.346, 9.7249, 9.4253, 8.9622, 9.507, 9.8475, 9.1937, 8.0088, 9.112, 9.7249, 9.2346, 9.5342, 10.038, 9.2346, 8.8124, 9.1529, 8.9622, 8.431, 8.7715, 9.0848, 8.6898, 8.8532, 8.8124, 8.8124, 8.54, 8.3084, 8.6625, 8.7306, 9.1937, 9.0439, 8.7306],
14     'time_step': [0, 0.030639, 0.069763, 0.099823, 0.12982, 0.15979, 0.18982, 0.2204, 0.24976, 0.27972, 0.31802, 0.34976, 0.37985, 0.40976, 0.44983, 0.4798, 0.50977, 0.53979, 0.56976, 0.59979, 0.62979, 0.65982, 0.68979, 0.72055, 0.75009, 0.78116, 0.80984, 0.83987, 0.86981, 0.89978, 0.92978, 0.9599, 0.98984, 1.0199,
```

```

14         1.0498, 1.0797, 1.1098, 1.1399, 1.1699, 1.1997, 1.2298, 1.2599,
15         1.2903, 1.3198, 1.3513, 1.3799, 1.4098, 1.4398, 1.4698, 1.4998]
16     })
17
18 # Separate features and labels
19 X = data[['time_step', 'x_acceleration', 'y_acceleration', '
20         z_acceleration']]
21 y = data['name']
22
23 # Encode labels
24 encoder = LabelEncoder()
25 y_encoded = encoder.fit_transform(y)
26
27 # Normalize features
28 scaler = StandardScaler()
29 X_scaled = scaler.fit_transform(X)
30
31 # Split the data
32 X_train, X_test, y_train, y_test = train_test_split(X_scaled,
33         y_encoded, test_size=0.2, random_state=42)
34
35 # Build the model
36 model = keras.Sequential([
37     keras.layers.Dense(64, activation='relu', input_shape=(X_train.
38         shape[1],)),
39     keras.layers.Dense(32, activation='relu'),
40     keras.layers.Dense(len(encoder.classes_), activation='softmax')
41 ])
42
43 # Compile the model
44 model.compile(optimizer='adam', loss='sparse_categorical_crossentropy',
45         metrics=['accuracy'])
46
47 # Train the model
48 history = model.fit(X_train, y_train, epochs=100, validation_data=(
49     X_test, y_test))
50
51 # Evaluate the model
52 test_loss, test_accuracy = model.evaluate(X_test, y_test)
53 print(f"Test Accuracy: {test_accuracy:.4f}")
54 print(f"Test Loss: {test_loss:.4f}")
55
56 # Generate classification metrics
57 from sklearn.metrics import classification_report
58 y_pred = model.predict(X_test)
59 y_pred_classes = np.argmax(y_pred, axis=1)
60 print("\nClassification Report:")
61 print(classification_report(y_test, y_pred_classes, target_names=
62     encoder.classes_))
63
64 # Save the trained model for integration with the Flutter app
65 model.save('footstep_identification_model.h5')
66 print("Model saved as 'footstep_identification_model.h5'")

```

Appendix B

Dart Code for Flutter Mobile App

```
1 // =====
2 // File: lib/main.dart
3 // Description: Main entry point for the app
4 // =====
5 import 'package:flutter/material.dart';
6 import 'home_page.dart';
7 import 'data_collection_page.dart';
8 import 'training_page.dart';
9 import 'prediction_page.dart';
10
11 void main() {
12   runApp(FootstepIdentificationApp());
13 }
14
15 class FootstepIdentificationApp extends StatelessWidget {
16   @override
17   Widget build(BuildContext context) {
18     return MaterialApp(
19       debugShowCheckedModeBanner: false,
20       title: 'Footstep Identification',
21       theme: ThemeData(
22         primarySwatch: Colors.blue,
23         fontFamily: 'Roboto',
24         scaffoldBackgroundColor: Colors.white,
25         textTheme: TextTheme(
26           bodyText1: TextStyle(fontSize: 16, color: Colors.black87),
27           bodyText2: TextStyle(fontSize: 14, color: Colors.black54),
28           headline6: TextStyle(fontSize: 20, fontWeight: FontWeight.
29             bold),
30         ),
31         elevatedButtonTheme: ElevatedButtonThemeData(
32           style: ElevatedButton.styleFrom(
33             padding: EdgeInsets.symmetric(vertical: 12, horizontal:
34               20),
35             textStyle: TextStyle(fontSize: 16, fontWeight: FontWeight.
36               bold),
37           ),
38         ),
39         initialRoute: '/',
40         routes: {
41           '/': (context) => HomePage(),
```

```

40         '/data-collection': (context) => DataCollectionPage(),
41         '/training': (context) => TrainingPage(),
42         '/prediction': (context) => PredictionPage(),
43     }
44 );
45 }
46 }
47
48 // =====
49 // File: lib/prediction_page.dart
50 // Description: UI for user prediction
51 // =====
52 import 'package:flutter/material.dart';
53
54 class PredictionPage extends StatelessWidget {
55   @override
56   Widget build(BuildContext context) {
57     String predictionResult = 'No prediction yet';
58     return Scaffold(
59       appBar: AppBar(title: Text('Prediction')),
60       body: Padding(
61         padding: const EdgeInsets.all(20.0),
62         child: Column(
63           mainAxisAlignment: MainAxisAlignment.center,
64           children: [
65             Text(
66               'Predict the individual based on footstep data.',
67               textAlign: TextAlign.center,
68               style: TextStyle(fontSize: 18),
69             ),
70             SizedBox(height: 20),
71             ElevatedButton.icon(
72               icon: Icon(Icons.upload_file),
73               label: Text('Upload Data'),
74               onPressed: () {
75                 // Simulate prediction process
76                 ScaffoldMessenger.of(context).showSnackBar(
77                   SnackBar(content: Text('Predicting...')),
78                 );
79               },
80             ),
81             SizedBox(height: 30),
82             Text(
83               'Result: $predictionResult',
84               style: TextStyle(fontSize: 18, fontWeight: FontWeight.
85                 bold),
86             ),
87             SizedBox(height: 30),
88             ElevatedButton(
89               onPressed: () {
90                 Navigator.pop(context);
91               },
92               child: Text('Back to Home'),
93             ),
94           ],
95         ),
96       );

```

```

97   }
98 }
99
100 // =====
101 // File: lib/training_page.dart
102 // Description: UI for initiating model training
103 // =====
104 import 'package:flutter/material.dart';
105
106 class TrainingPage extends StatefulWidget {
107   @override
108   _TrainingPageState createState() => _TrainingPageState();
109 }
110
111 class _TrainingPageState extends State<TrainingPage> {
112   bool isTraining = false;
113
114   void startTraining() {
115     setState(() {
116       isTraining = true;
117     });
118     // Simulate a delay for training process
119     Future.delayed(Duration(seconds: 5), () {
120       setState(() {
121         isTraining = false;
122       });
123       ScaffoldMessenger.of(context).showSnackBar(
124         SnackBar(content: Text('Model trained successfully!')),
125       );
126     });
127   }
128
129   @override
130   Widget build(BuildContext context) {
131     return Scaffold(
132       appBar: AppBar(title: Text('Train Model')),
133       body: Padding(
134         padding: const EdgeInsets.all(20.0),
135         child: Column(
136           mainAxisAlignment: MainAxisAlignment.center,
137           children: [
138             Text(
139               'Train your model with the collected data.',
140               style: TextStyle(fontSize: 18),
141               textAlign: TextAlign.center,
142             ),
143             SizedBox(height: 30),
144             if (isTraining)
145               Column(
146                 children: [
147                   CircularProgressIndicator(),
148                   SizedBox(height: 10),
149                   Text('Training in progress...'),
150                 ],
151               )
152             else
153               ElevatedButton.icon(
154                 icon: Icon(Icons.play_circle_fill),

```

```

155         label: Text('Start Training'),
156         onPressed: startTraining,
157     ),
158     SizedBox(height: 30),
159     ElevatedButton(
160         onPressed: () {
161             Navigator.pop(context);
162         },
163         child: Text('Back to Home'),
164     ),
165 ],
166 ),
167 ),
168 );
169 }
170 }
171
172 // =====
173 // File: lib/data_collection_page.dart
174 // Description: UI for capturing sensor data
175 // =====
176 import 'package:flutter/material.dart';
177
178 class DataCollectionPage extends StatelessWidget {
179   @override
180   Widget build(BuildContext context) {
181     bool isCollecting = false; // Simulating sensor status
182     return Scaffold(
183       appBar: AppBar(title: Text('Data Collection')),
184       body: Padding(
185         padding: const EdgeInsets.all(20.0),
186         child: Column(
187           mainAxisAlignment: MainAxisAlignment.center,
188           children: [
189             Text(
190               'Connect your sensors to start collecting data.',
191               style: TextStyle(fontSize: 18, color: Colors.grey[800]),
192               textAlign: TextAlign.center,
193             ),
194             SizedBox(height: 20),
195             Card(
196               color: isCollecting ? Colors.green[100] : Colors.red
197               [100],
198               shape: RoundedRectangleBorder(
199                 borderRadius: BorderRadius.circular(10),
200               ),
201               elevation: 4,
202               child: ListTile(
203                 leading: Icon(
204                   isCollecting ? Icons.check_circle : Icons.error,
205                   color: isCollecting ? Colors.green : Colors.red,
206                 ),
207                 title: Text(
208                   isCollecting ? 'Sensor Connected' : 'No Sensor
209 Detected',
210                   style: TextStyle(fontSize: 16, fontWeight:

```

```

210         ),
211     ),
212     SizedBox(height: 30),
213     ElevatedButton.icon(
214         icon: Icon(Icons.sensors),
215         label: Text('Start Data Collection'),
216         onPressed: () {
217             ScaffoldMessenger.of(context).showSnackBar(
218                 SnackBar(content: Text('Data collection started...'))
219             ),
220         );
221     },
222     SizedBox(height: 20),
223     ElevatedButton(
224         onPressed: () {
225             Navigator.pop(context);
226         },
227         child: Text('Back to Home'),
228     ),
229 ],
230 ),
231 ),
232 );
233 }
234 }
235
236 // =====
237 // File: lib/home_page.dart
238 // Description: Landing page with navigation
239 // =====
240 import 'package:flutter/material.dart';
241
242 class HomePage extends StatelessWidget {
243   @override
244   Widget build(BuildContext context) {
245     return Scaffold(
246       appBar: AppBar(title: Text('Footstep Identification')),
247       body: Padding(
248         padding: const EdgeInsets.all(20.0),
249         child: Column(
250           mainAxisAlignment: MainAxisAlignment.center,
251           children: [
252             Text(
253               'Welcome!',
254               style: TextStyle(fontSize: 28, fontWeight: FontWeight.
255 bold),
256             ),
257             SizedBox(height: 10),
258             Text(
259               'Identify individuals by their unique footsteps using
260 advanced ML.',
261               textAlign: TextAlign.center,
262               style: TextStyle(fontSize: 16, color: Colors.grey[700]),
263             ),
264             SizedBox(height: 30),
265             Expanded(
266               child: GridView.count(

```

```

265         crossAxisCount: 2,
266         crossAxisSpacing: 10,
267         mainAxisSpacing: 10,
268         children: [
269             _buildMenuCard(
270                 context,
271                 title: 'Collect Data',
272                 icon: Icons.sensors,
273                 route: '/data-collection',
274             ),
275             _buildMenuCard(
276                 context,
277                 title: 'Train Model',
278                 icon: Icons.bar_chart,
279                 route: '/training',
280             ),
281             _buildMenuCard(
282                 context,
283                 title: 'Predict',
284                 icon: Icons.insights,
285                 route: '/prediction',
286             ),
287         ],
288     ),
289 ),
290 ],
291 ),
292 ),
293 );
294 }
295
296 Widget _buildMenuCard(BuildContext context, {required String title,
297   required IconData icon, required String route}) {
298   return GestureDetector(
299     onTap: () {
300       Navigator.pushNamed(context, route);
301     },
302     child: Card(
303       shape: RoundedRectangleBorder(borderRadius: BorderRadius.
304         circular(15)),
305       elevation: 4,
306       child: Column(
307         mainAxisAlignment: MainAxisAlignment.center,
308         children: [
309           Icon(icon, size: 40, color: Colors.blue),
310           SizedBox(height: 10),
311           Text(
312             title,
313             textAlign: TextAlign.center,
314             style: TextStyle(fontSize: 16, fontWeight: FontWeight.
315               w600),
316           ),
317         ],
318       ),
319     );

```